Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/25/2007
PI Name:	Bronner-Fraser, Marianne Ph.D.		
Project Title:	Evolution Of The Neural Crest And Its Contribution To The Inner Ear		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Evolutionary biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
PI Email:	mbronner@gg.caltech.edu	**Fax:**	FY 626-395-7717
PI Organization Type:	UNIVERSITY	**Phone:**	626-395-3355
Organization Name:	California Institute of Technology		
PI Address 1:	Division of Biology, Mail Stop 139-74		
PI Address 2:	1200 East California Boulevard		
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91125	**Congressional District:**	29
Comments:	B		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: on NCE to 10/31/2006, per A. Chu/ARC (4/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Holland, Linda (Scripps Oceanographic Institute)		
Grant/Contract No.:	NAG2-1585		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The neural crest and the cranial ectodermal placodes give rise to many of the defining characters of vertebrates, including a well-defined head with teeth and paired sensory organs. Both are migratory, form at the border between neural plate and epidermis, and generate some identical derivatives such as sensory neurons. To explore the events that led to the evolution of vertebrate features, this research will examine: (1) the evolutionary origin of the neural crest and placodes and (2) the interrelationship and evolutionary divergence of these two important embryonic structures. We aim to look for evolutionary precursors to neural crest and placodes in the invertebrate chordate, amphioxus, and to study neural crest and placodes in the primitive vertebrate, lamprey. We will utilize vital dye labeling to examine whether amphioxus has migratory cells that resemble neural crest/placodal cells in vertebrates. We will correlate the sources of migratory cells with gene expression domains of homologs to vertebrate molecular markers for neural crest cells and placodes. In addition, we will map neural crest migratory pathways and derivatives in the lamprey. These represent questions of major importance in the evolution of the vertebrate body plan. Task Objective: Examine evolutionary origin of vertebrate neural crest and placodes, with emphasis on ear. Task Significance: Important for understanding vertebrate origins as well as evolution of vestibular system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the origin of life on earth is critical to understanding if life may have evolved elsewhere in the universe. This research promises to be highly informative regarding the origin of vertebrate body plan. Neural crest cells and placodes are unique to vertebrates and are some of their most defining characteristics. However, little is known about how these tissue types evolved. By understanding their evolutionary origin from non-vertebrate chordates, we hope to better discern how the vertebrate body plan arose.
Task Progress:	NO PROGRESS MADE DURING THE LAST YEAR OF THIS GRANT BY PI BRONNER-FRASER. BELOW REPORT IS FOR CO-INVESTIGATOR LINDA HOLLAND: During the no-cost extension, this grant continued to support a graduate student Jr-Kai Yu, who received his Ph. D. late in 2005. Dr. Yu extended his investigations into the forkhead gene family, cloning amphioxus homologs of FoxN and FoxI and determining their expression during amphioxus embryogenesis. Moreover, in a collaboration with the laboratory of N. Satoh in Japan, Dr. Yu participated in a large-scale cDNA project that resulted in end-sequencing of 210,000 clones. From these sequences, Dr. Yu identified 23 genes that were homologs of vertebrate genes involved in early dorso/ventral (D/V) and anterior/posterior (A/P) patterning. This work showed that the gastrula of the cephalochordate amphioxus expresses D/V patterning genes (e.g. BMPs, Nodal and their antagonists) in patterns reminiscent of those of their vertebrate orthlogues, and that amphioxus embryos, like those of vertebrates are ventralized by exogenous BMP protein. In addition, Wnt-antagonists (e.g. Dkks, sFRP2-like) are expressed anteriorly, while Wnt genes themselves are expressed posteriorly, consistent with a role for Wnt signaling A/P patterning. These results suggested evolutionary conservation of the mechanisms for both D/V and A/P patterning of the early gastrula in amphioxus and vertebrates. In light of recent phylogenetic analyses placing cephalochordates basally in the chordate lineage, we propose that separate signaling centers for patterning the D/V and A/P axes may be an ancestral chordate character. This work was published in Nature (Yu JK, Satou Y, Holland ND, Shin-I T, Kohara Y, Satoh N, Bronner-Fraser M, Holland LZ. Axial patterning in cephalochordates and the evolution of the organizer. Nature. 2007 Feb 8;445(7128):613-7). In addition, Dr. Yu identified numerous amphioxus homologs of vertebrate neural crest genes amongst the 210,000 clones and has been continuing this project in his current postdoctoral position.
Bibliography Type:	Description: (Last Updated: 05/04/2007)
Articles in Peer-reviewed Journals	Yu JK, Satou Y, Holland ND, Shin-I T, Kohara Y, Satoh N, Bronner-Fraser M, Holland LZ. "Axial patterning in cephalochordates and the evolution of the organizer." Nature. 2007 Feb 8;445(7128):613-7. PMID: 17237766 , Feb-2007